OUR VERY PERSONAL
(AND NON-PEER REVIEWED)

Survival guide to **programming** for Women+ in Psychology

arQus European University Alliance

cimcyc

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UNIVERSIDAD DE GRANADA

UNIVERSITÀ DEGLI STUDI DI PADOVA

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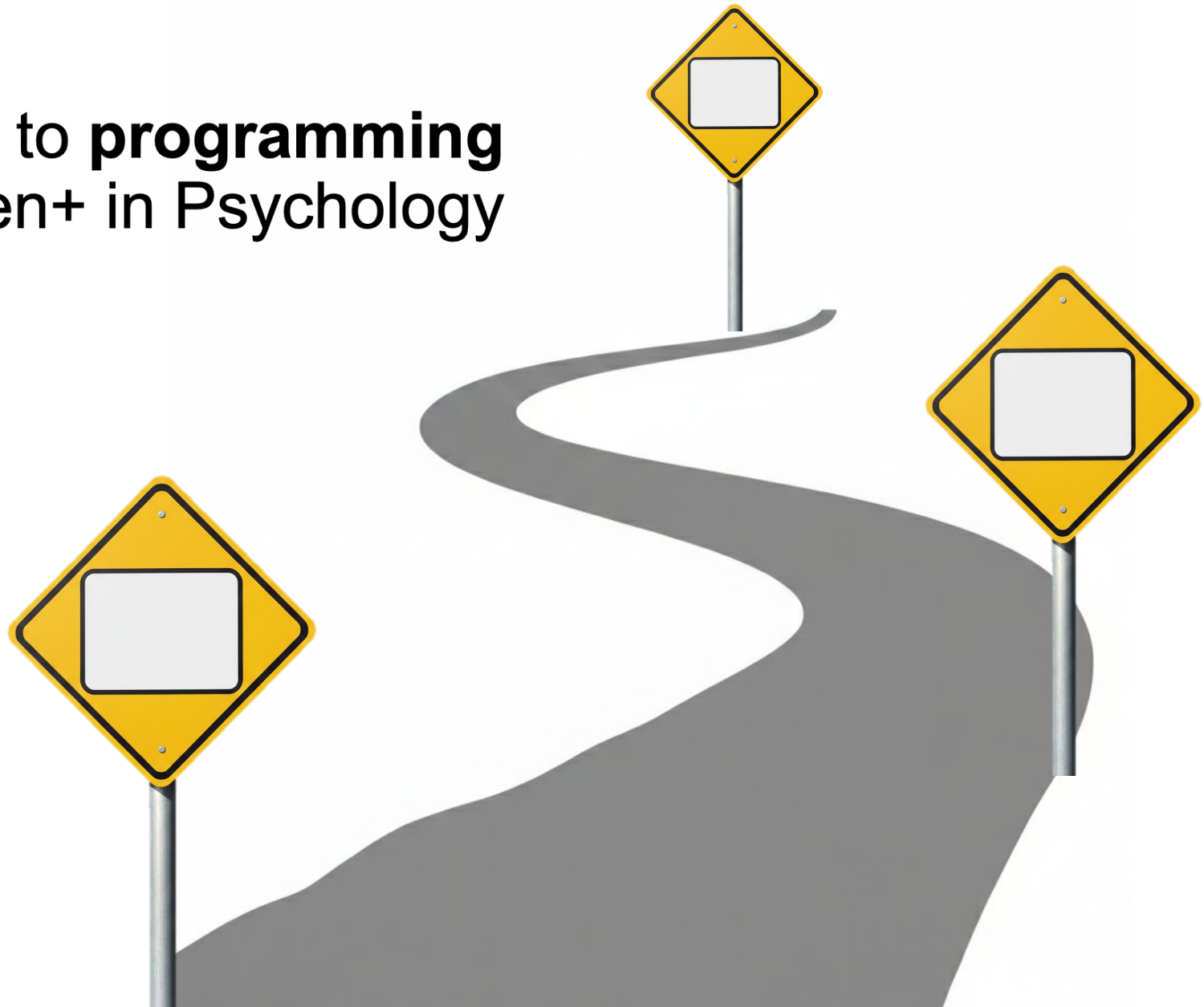

Do I *really need* to learn how to program as a psychologist?

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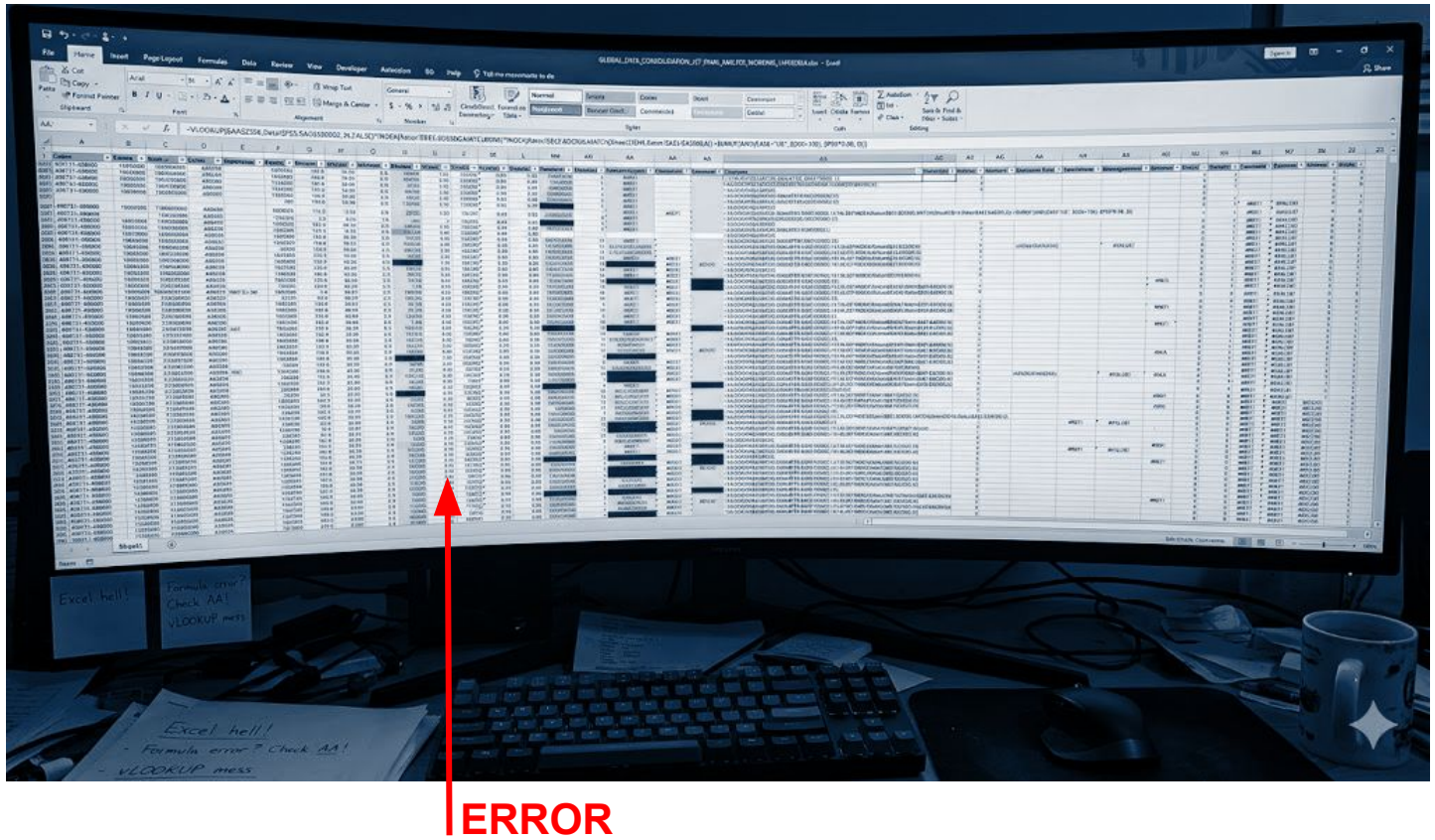

ERROR

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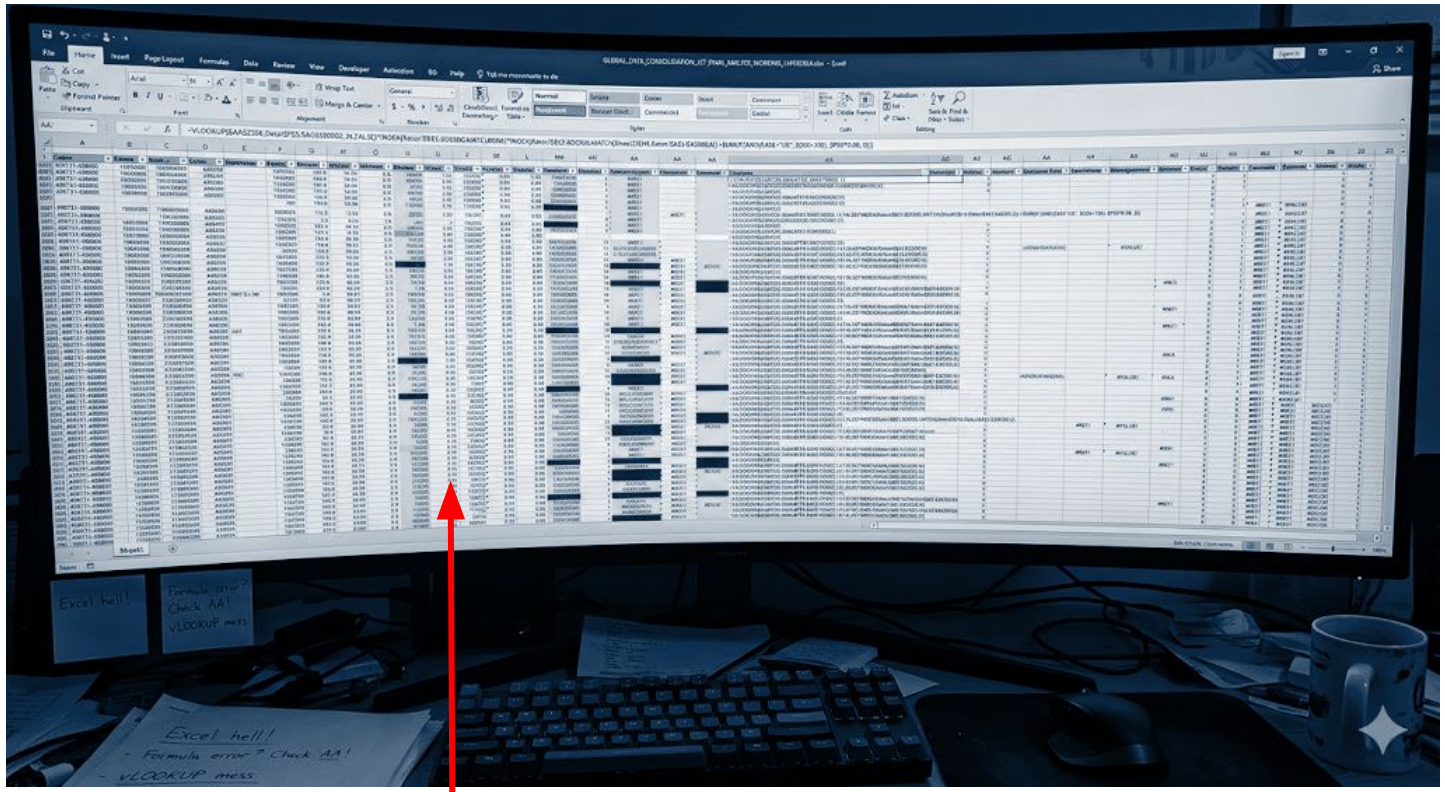

ERROR

For me… 😌

CONTROL

For others… 🫂

PRECISION 💪

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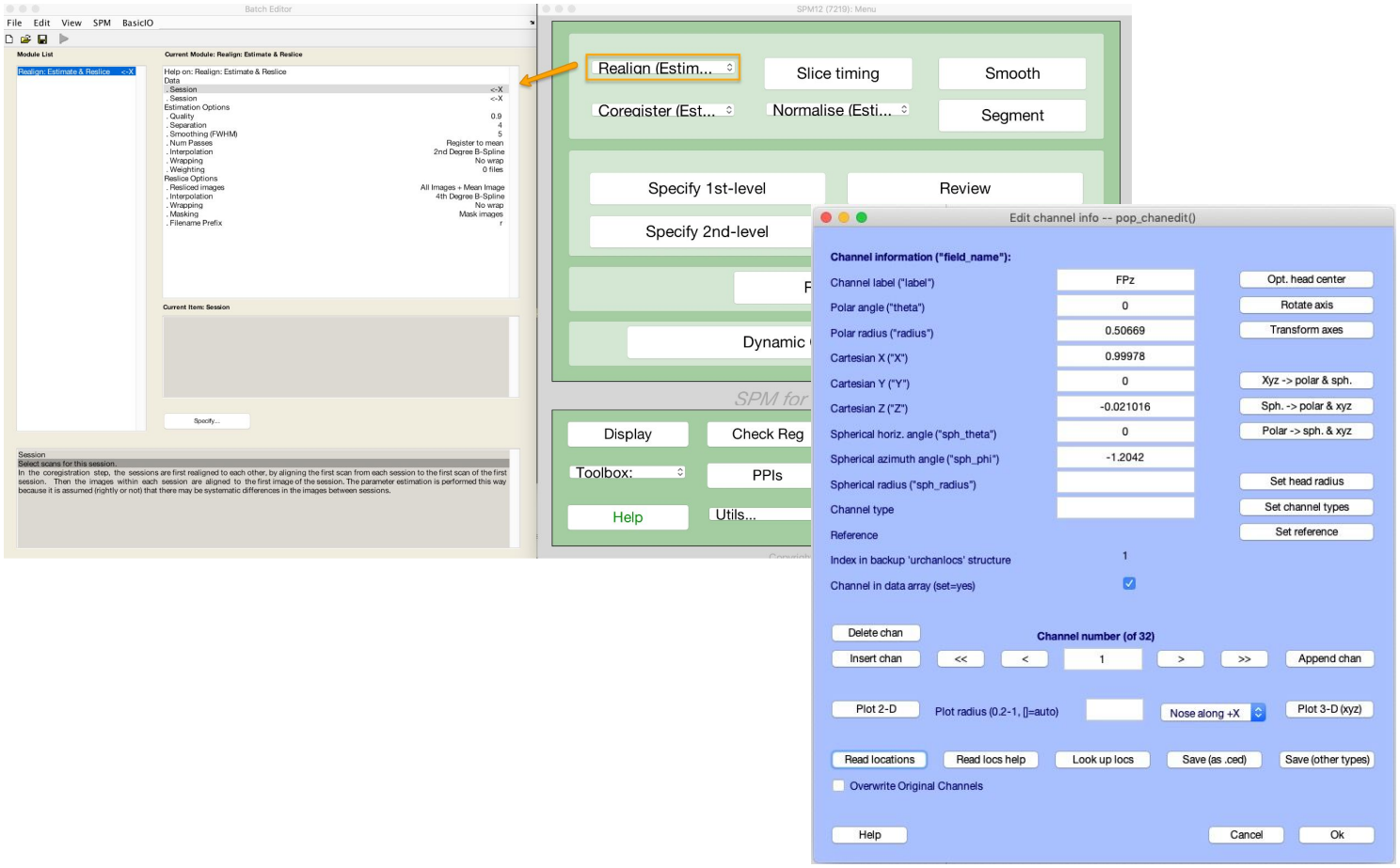

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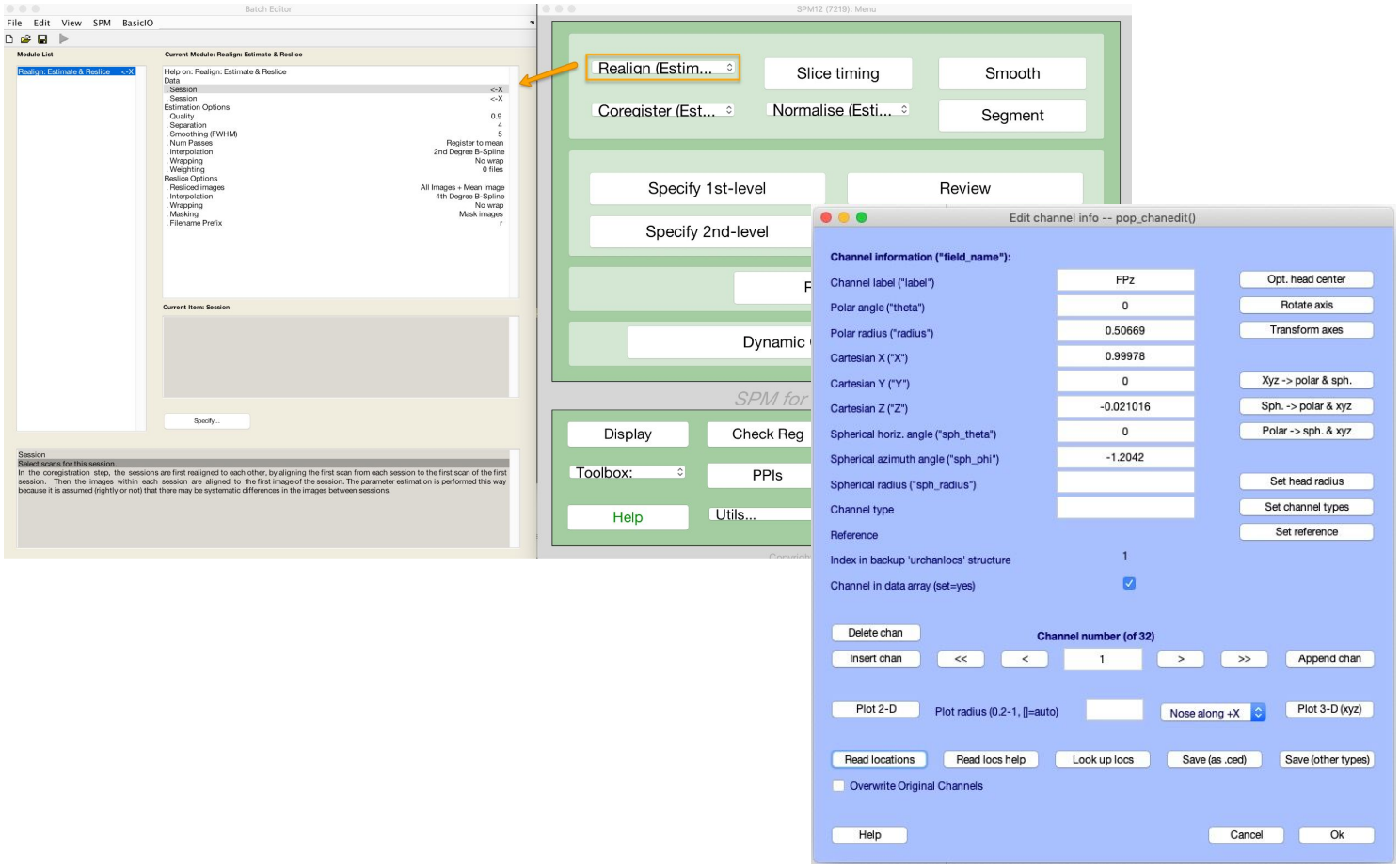

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CONTROL

FLEXIBILITY

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PRECISION 💪

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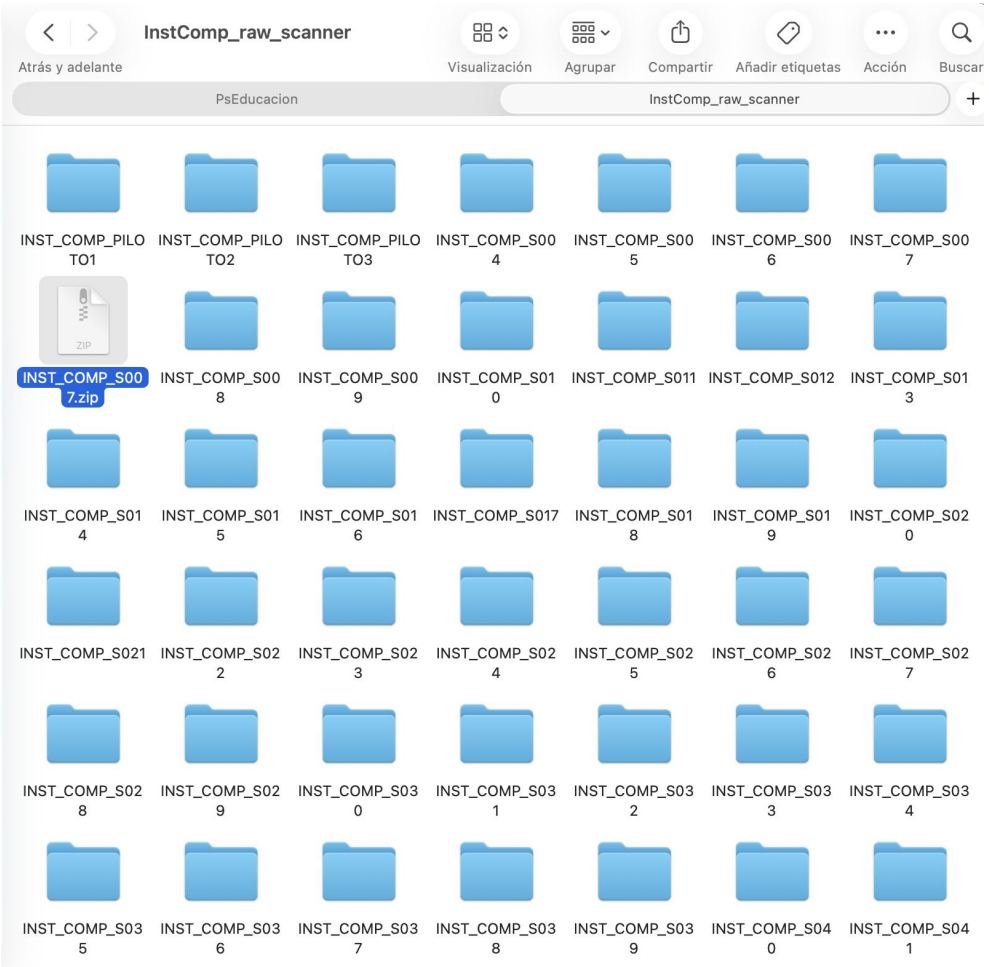


x 100 subjects

For me… 😌

CONTROL

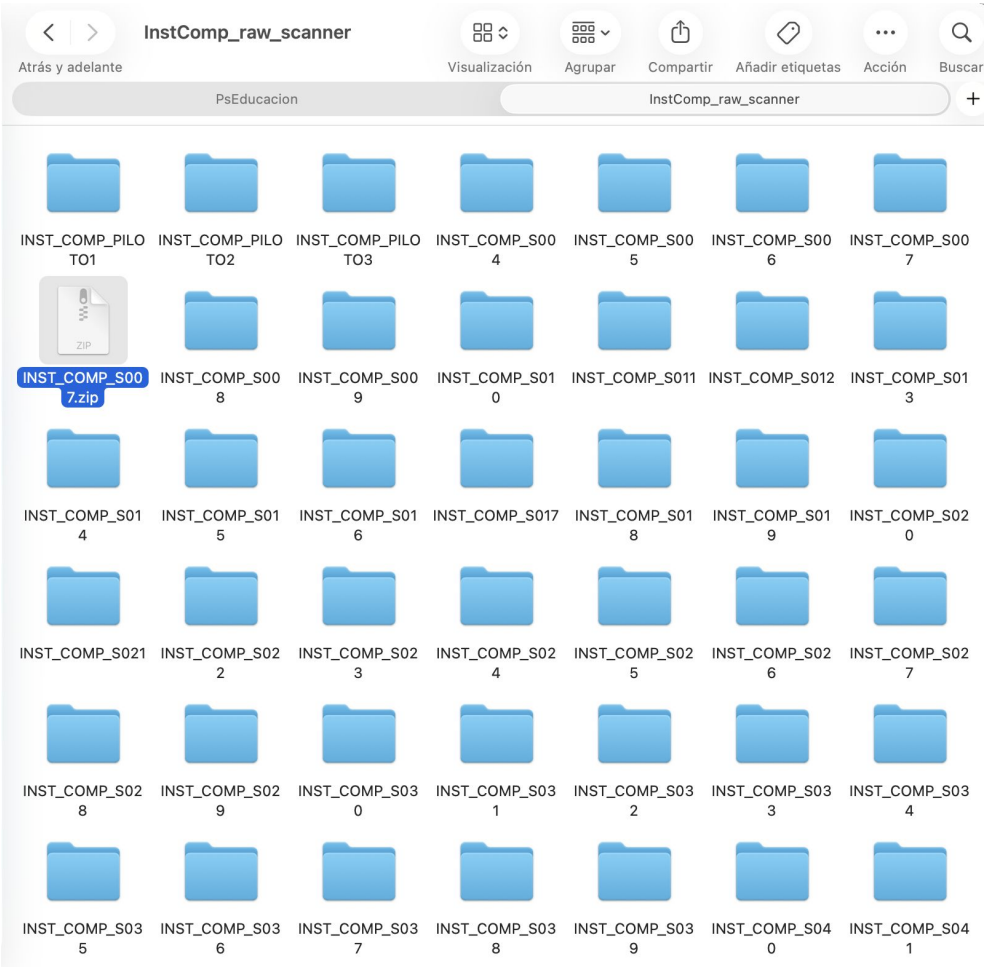
FLEXIBILITY

For others… 🫂

PRECISION 💪

TRANSPARENCY 👀

Do I *really need* to learn how to program as a psychologist?



x 100 subjects

For me… 😌

CONTROL

FLEXIBILITY

AUTOMATICITY

For others… 🫂

PRECISION 💪

TRANSPARENCY 👀

REPRODUCIBILITY ✨

Do I *really need* to learn how to program as a psychologist?

CODING IS A POWERFUL, HIGHLY VALUED TECHNICAL SKILL
that enhances students' employability both inside and outside academia

For me… 😌

CONTROL
FLEXIBILITY
AUTOMATICITY

For others… 🫂

PRECISION 💪
TRANSPARENCY 👀
REPRODUCIBILITY ✨

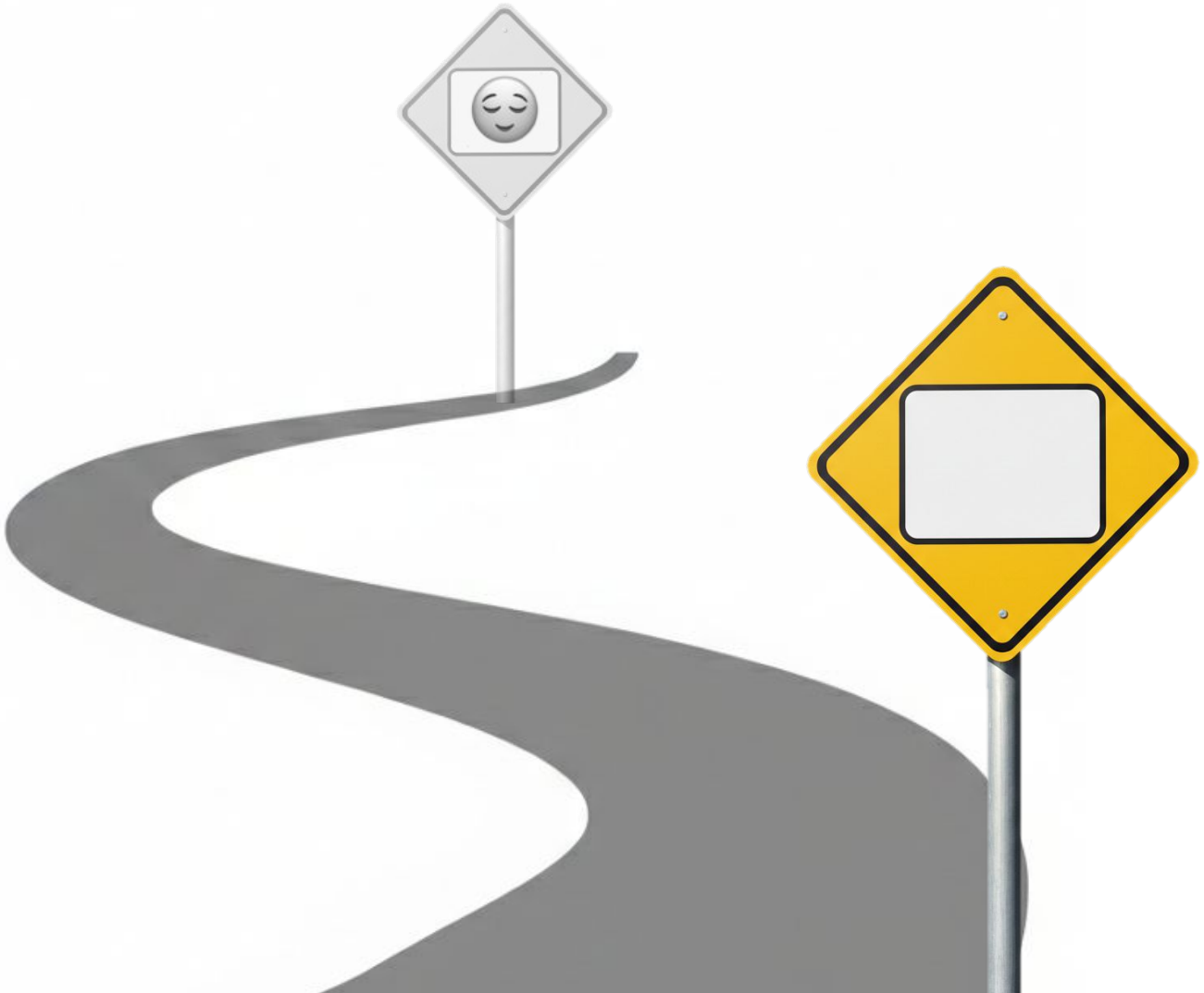
Do I *really need* to learn how to program as a psychologist?

- Control, flexibility and automaticity.
- Precision, transparency and reproducibility

- Better job prospects

Definitely <u>YES!</u>

Why is it *so hard* to access programing as women+ from the field of psychology?



“Back then, programming felt like mysterious magic that I just couldn’t access to”

My friend Mayte: an incredibly talented researcher (female & psychologist) who codes.

When we think about programmers, developers, computer scientists…

- Male, probably white
- ***Brilliant***, great intelligence
- Socially awkward
- Narrow, focused interests
- Nerd / Geek

[Cheryan et al., 203; Pantic et al., 2018]

Please draw a highly skilled professional programmer.

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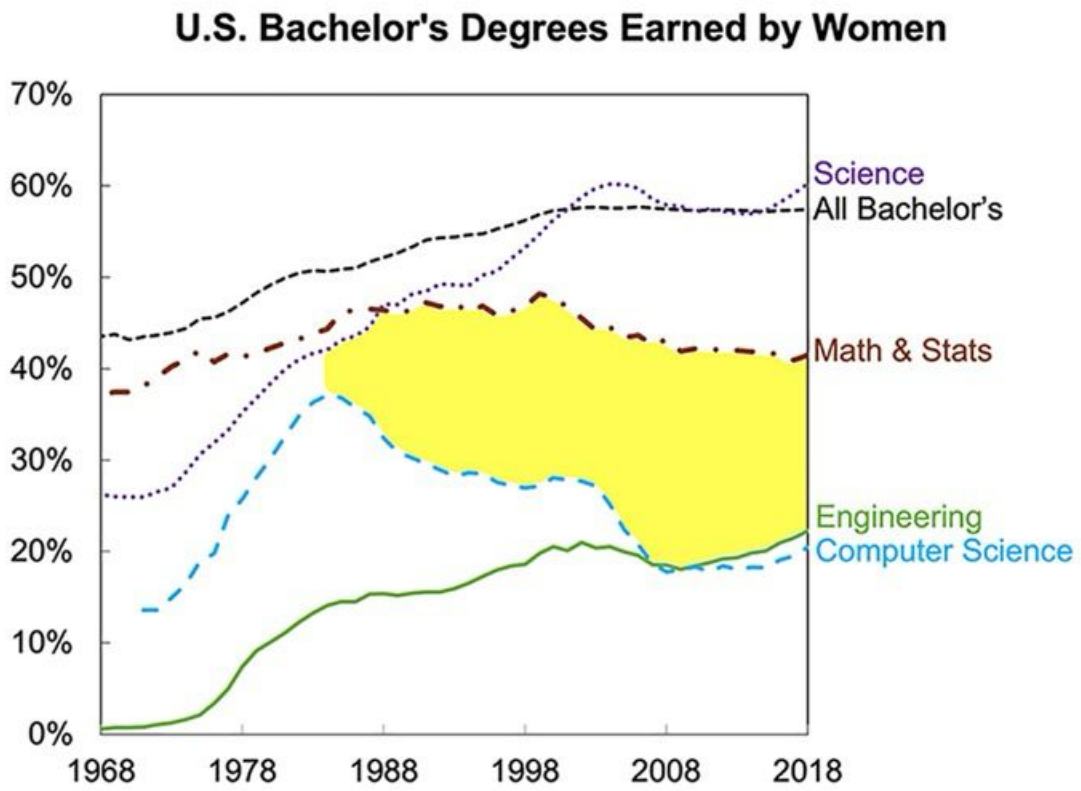
[Cheryan et al., 203; Pantic et al., 2018]

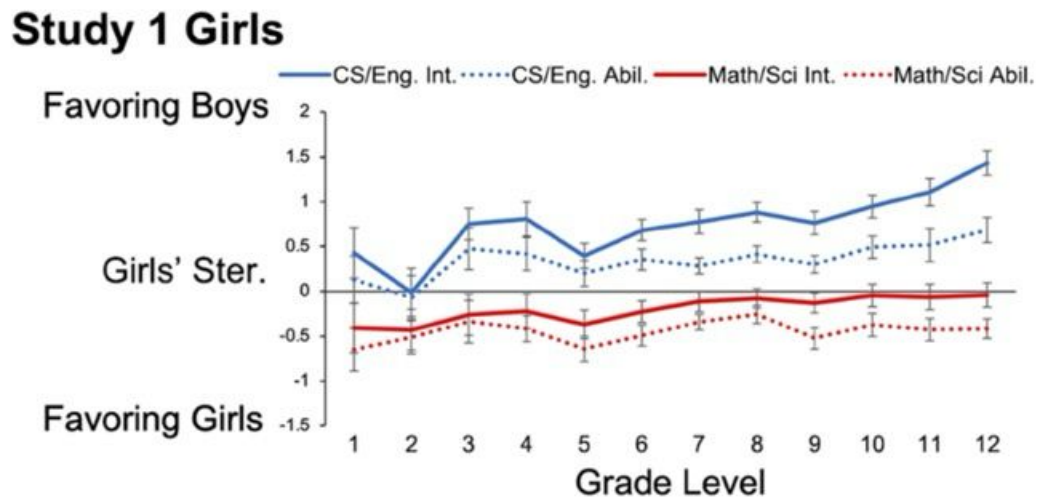
The **stereotypical, male-dominated view** of computer science **exceeds** other STEM domains and **emerges early** (6-8 years old) [e.g., Miller et al., 2024; Master et al., 2025]

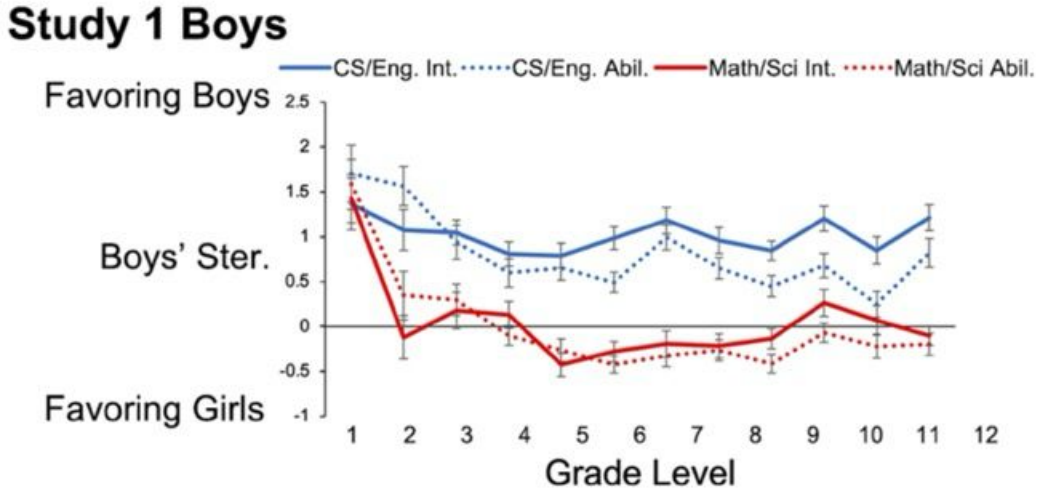
PNAS RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES

Divergence in children's gender stereotypes and motivation across STEM fields

Allison Master[a,1], Andrew N. Meltzoff[b,c], Daijiazi Tang[a,d], and Sapna Cheryan[c]







When we think about psychologists or psychology…

Woman!
Extremely feminized field

Prosocial, empathetic, care-giver

Soft science (or not a science), overall *easy*

[Boysen et al., 2021; Brinthaupt et al., 2016; Alperin et al., 1985]

Please draw a highly skilled professional psychologist.

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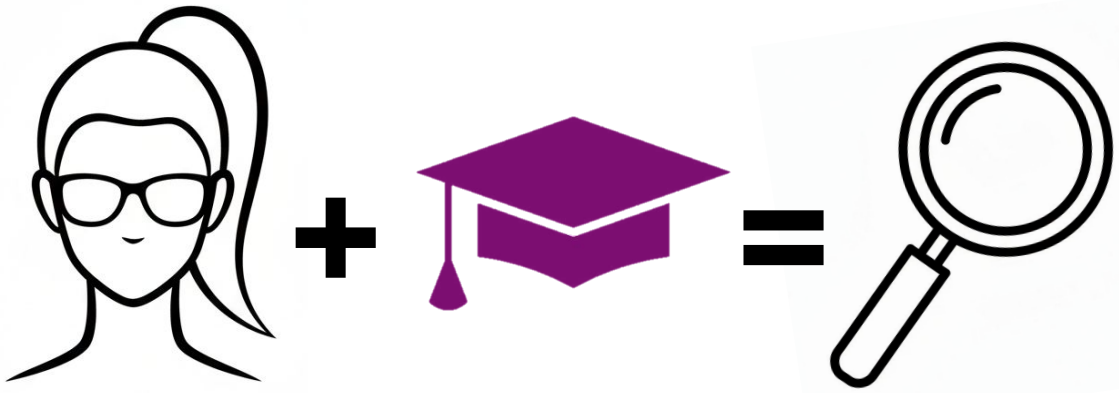
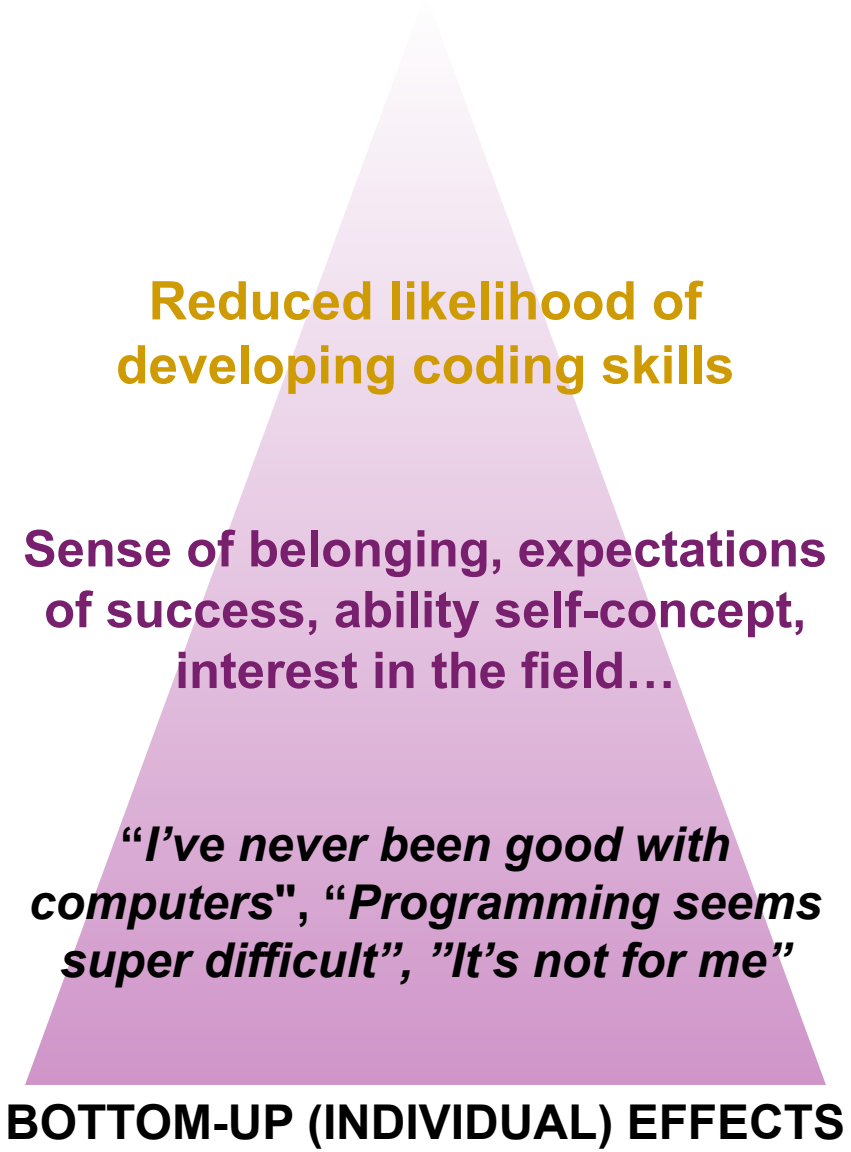


Python

Python
COLLIDING
STEREOTYPES


Reduced likelihood of developing coding skills
Sense of belonging, expectations of success, ability self-concept, interest in the field…
“I’ve never been good with computers", “Programming seems super difficult”, ”It’s not for me”
BOTTOM-UP (INDIVIDUAL) EFFECTS

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BOTTOM-UP (INDIVIDUAL) EFFECTS

TOP-DOWN (STRUCTURAL) EFFECTS

"Psychologist won't learn easily how to program", "Coding is useless for a psychologist"

At many universities, programming is not part of the Psychology curriculum.

Psychology students don't get expose to programming

Why is it *so hard* to access programing as a women in cognitive science?

- Pervasive gender gap in COMPUTATIONAL skills
- Polarized, colliding stereotypes!

Socially shared (and perpetuated) stereotypes shape decision-making processes both at the individual and institutional levels.

WHAT *can we do* to foster access to programming from psychology?

1. **Early exposure to programming.**

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2. **Do not ignore the elephant in the room! Let's go *beyond coding* and talk about:**
   - **the gender gap,**
   - **stereotypes & stereotype threat,**
   - **counterstereotypical examples, …**

Image source: Wikimedia Commons

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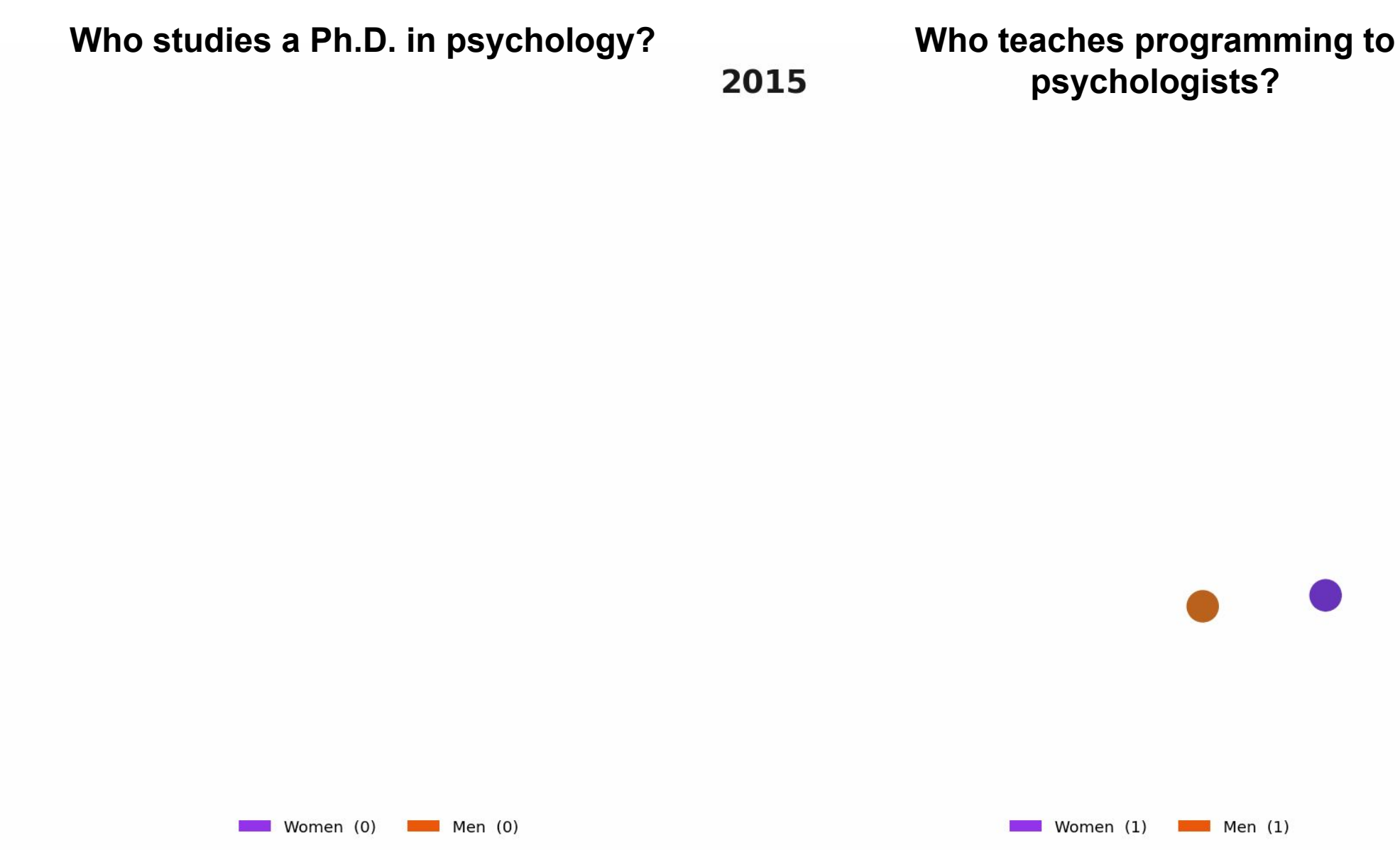
3. **Diversify role models.**

Who studies a Ph.D. in psychology?

Who teaches programming to psychologists?

Data from the **University of Granada** Ph.D. Program in Psychology and the Mind, Brain and Behavior Research Center.



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   - **counterstereotypical examples, …**
3. **Diversify role models.**
4. **Fight barriers widening the *real* access.**

Programming in Psychological Science A Practical Introduction

Start reading! View on GitHub

https://wobc.github.io/programming_book/

This guide is **not** a full programming course or a replacement for formal tutorials. Instead, it's a **starting point**: a collection of reflections, examples, and recommendations from the perspective of researchers in psychology and cognitive neuroscience. It aims to help you get your bearings and make smart decisions as you begin learning to code — not to teach you everything from scratch.

Chiara Avancini

Luis Ciria

Carlos González-García

Juan Linde-Domingo

Maria Jesús Maraver

Javier Ortiz-Tudela

Ana F. Palenciano

Rodika Sokoliuk

neuroplausible About

Why Women in Psychology Can't Program

Olivia Guest November 26, 2018 18 Comments

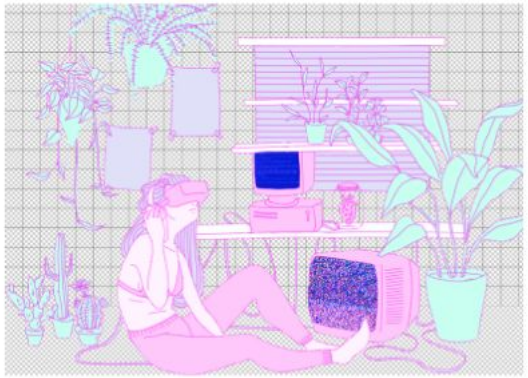
Share Tweet

They could if given the chance, of course! Sadly basic coding skills are being gatekept due to elitism and sexism from the teaching side with arguments along the lines of: "they just cannot cope with learning how to code".

About two months ago my brother, who works in data science on social psychology data, asked me why his colleagues, who are women and have PhDs in psychology, cannot code and why they use SPSS. He was obviously just venting because when I replied he was surprised. I told him that it was because of sexism and because of lack of proper skills-training (both for statistics and coding) in many psychology departments. That got me thinking about this more…

Why can't they code? Because they aren't taught how to code. *Why aren't they taught?* Because of two main, related, reasons:

- Rhetoric and pressure from formal societies like the BPS as well as gatekeepers who worry about how the field might/could split as it is true that a undergrad degree (especially) is essentially zero sum. We just can't fit it all in and coding is seen as not that relevant or worthy of being inserted into the curriculum.
- Researchers and educators have internalised ideas about what undergrads, who are predominantly women, can and cannot learn. And



"[That feeling when] your plant friends die cause you spend way too much time in virtual realities" by écharpe

Amsterdam University Press

TIJDSCHRIFT VOOR GENDERSTUDIES 27.2/3 (2024) 196-217
HTTPS://DOI.ORG/10.5117/TVGN2024.2-3.007.GUES

Teaching coding inclusively

If this, then what?

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Samuel H. Forbes
Department of Psychology, Durham University
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Olivia Guest:https://neuroplausible.com/programming

https://doi.org/10.31235/osf.io/3r2ez

Thanks for listening, *and let's do this!* 💪

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EXCELENCIA MARÍA DE MAEZTU

UNIVERSIDAD DE GRANADA

UNIVERSITÀ DEGLI STUDI DI PADOVA

Education and skills

Graduates %	Female	Male	Value
STEM	10.45	35.46	21.34
Agri., Forestry, Fisheries & Veterinary	0.89	1.22	1.03
Arts & Humanities	9.13	8.04	8.65
Business, Admin. & Law	19.30	19.25	19.28
Education	20.58	9.89	15.92
Engineering, Manuf. & Construction	5.22	19.58	11.47
Health & Welfare	24.76	10.78	18.67
Information & Comm. Technologies	1.26	10.38	5.23
Natural Sci., Mathematics & Statistics	3.97	5.50	4.64
Social Sci., Journalism & Information	8.88	5.75	7.52